

Numerical Exercise #4

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December 20, 2025

1 Question #1 Initial Loading Pattern

keff (dimensionless, 5 decimal points):

1.21882

peak power (relative to average, dimensionless, 2 decimal points):

2.44

vessel fast flux (2 decimal points):

$7.23 \times 10^{13} \text{ n cm}^{-2} \text{ s}^{-1}$

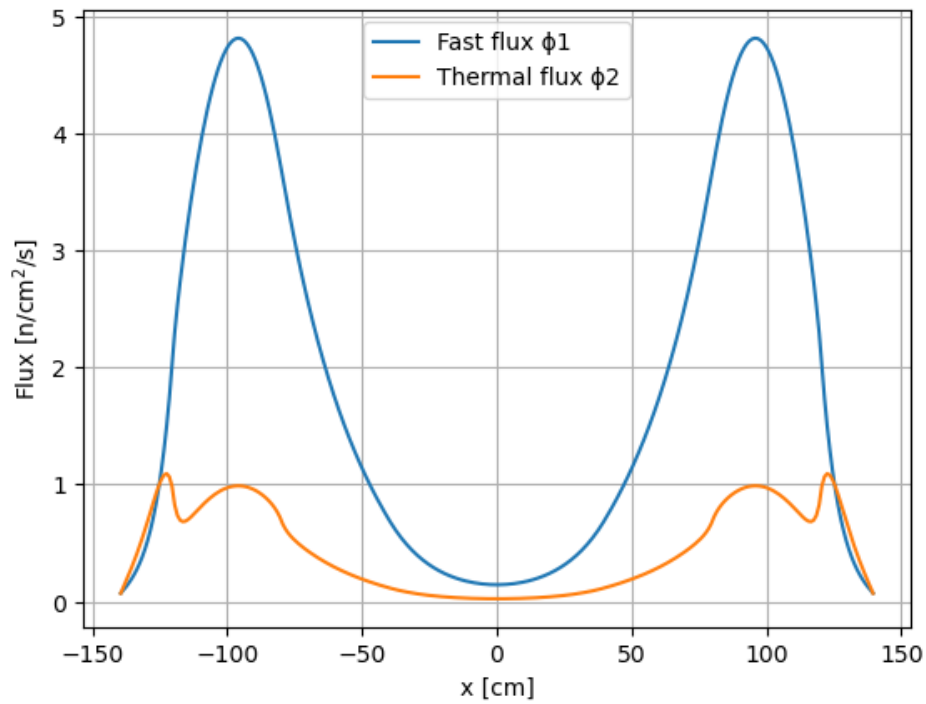


Figure 1: Normalized fast and thermal fluxes

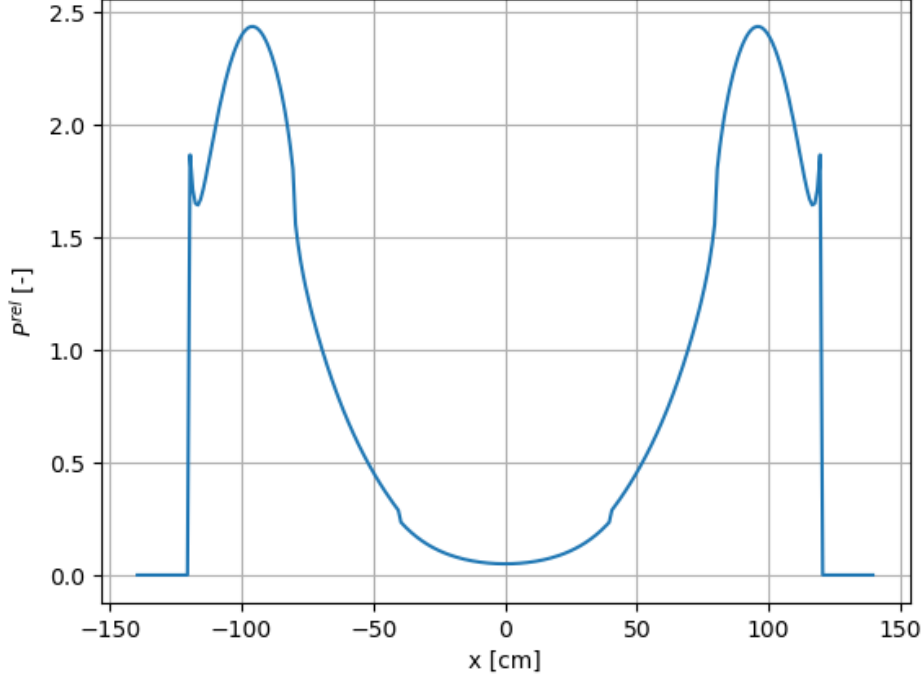


Figure 2: Relative power distribution

Explanation of the main features of the power distribution:

The relative power distribution is symmetric with respect to the core center, as expected from the 1D slab model and the symmetry boundary condition at $x = 0$. Power is produced only in the fuel region because $P_i \propto \sum_g \kappa \Sigma_{f,g} \phi_g$ and $\kappa \Sigma_f = 0$ in the reflector; therefore P^{rel} drops to zero outside the core.

The profile exhibits pronounced edge peaking near the core/reflector interfaces. This is mainly caused by the reflector returning neutrons back into the core, which increases the local thermal flux close to the boundary and enhances fission in the thermal group. In the present loading pattern [3 3 2 2 1 1] (from center to periphery), the freshest fuel (ID 1) is located at the outer part of the core, where both the neutron flux is boosted by the reflector and the macroscopic fission terms are higher, leading to the observed maxima of P^{rel} near the periphery.

Finally, small slope changes are visible at assembly interfaces because cross sections are piecewise constant by assembly (IDs $3 \rightarrow 2 \rightarrow 1$), which locally modifies diffusion, absorption and scattering properties and thus changes the curvature of the flux and power profiles.

2 Question #2 Optimized Loading Pattern

1	1	1	1	1	1	4
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Table 1: Loading pattern from the core center (left to right)

k_{eff} (dimensionless, 5 decimal points):

1.260 19

peak power (relative to average, dimensionless, 2 decimal points):

2.99

vessel fast flux (2 decimal points):

$1.54 \times 10^{12} \text{ n cm}^{-2} \text{ s}^{-1}$

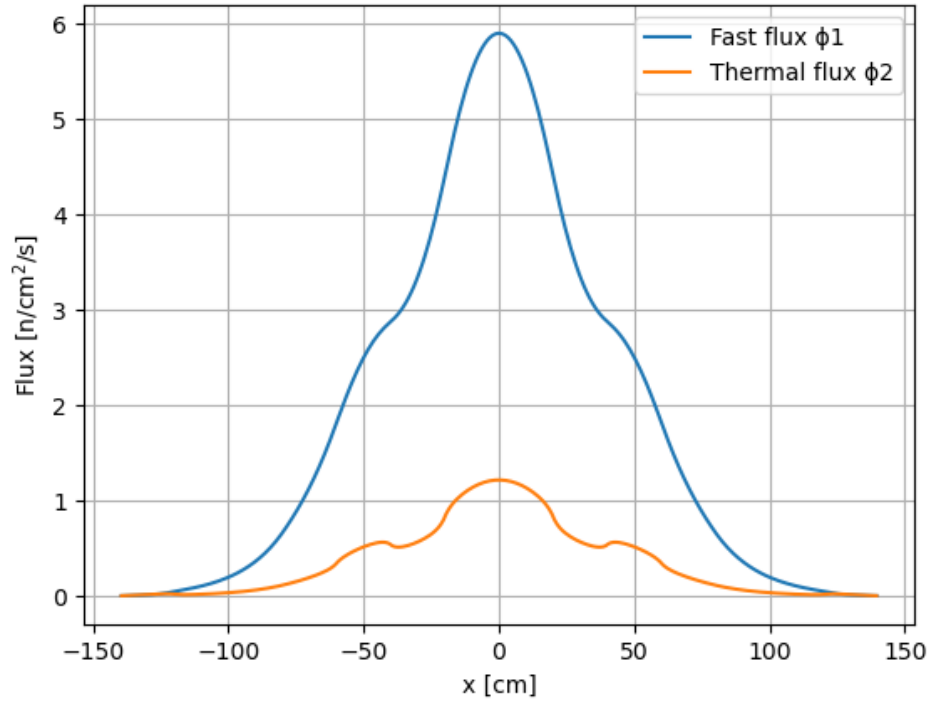


Figure 3: Normalized fast and thermal fluxes

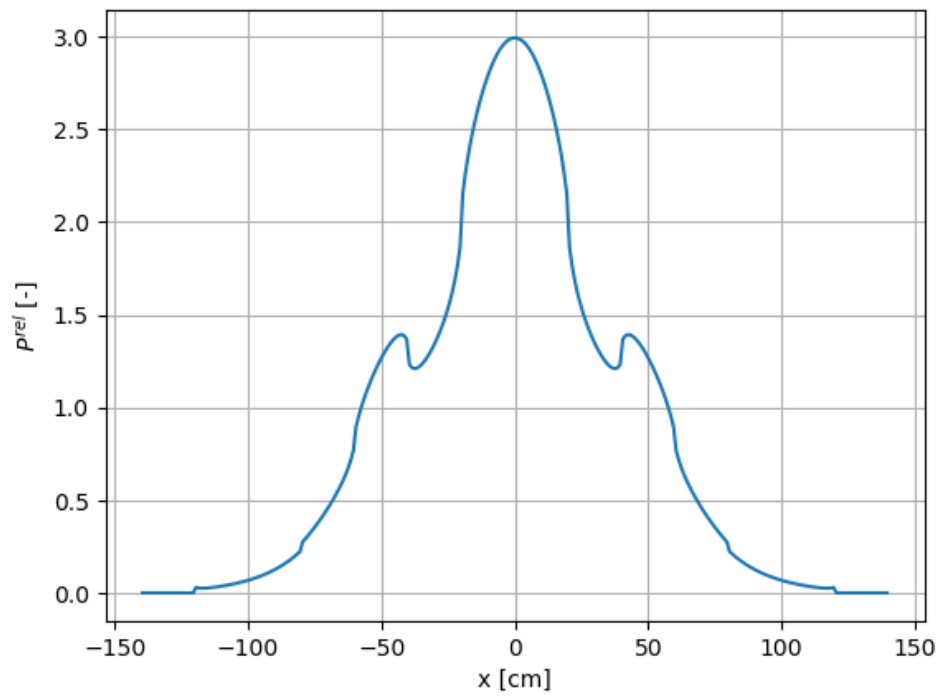


Figure 4: Relative power distribution